

UNIVERSIDAD
DE MÁLAGA

Sistemas en Tiempo Real

Seminario

ATMEGA328p/ARDUINO UNO

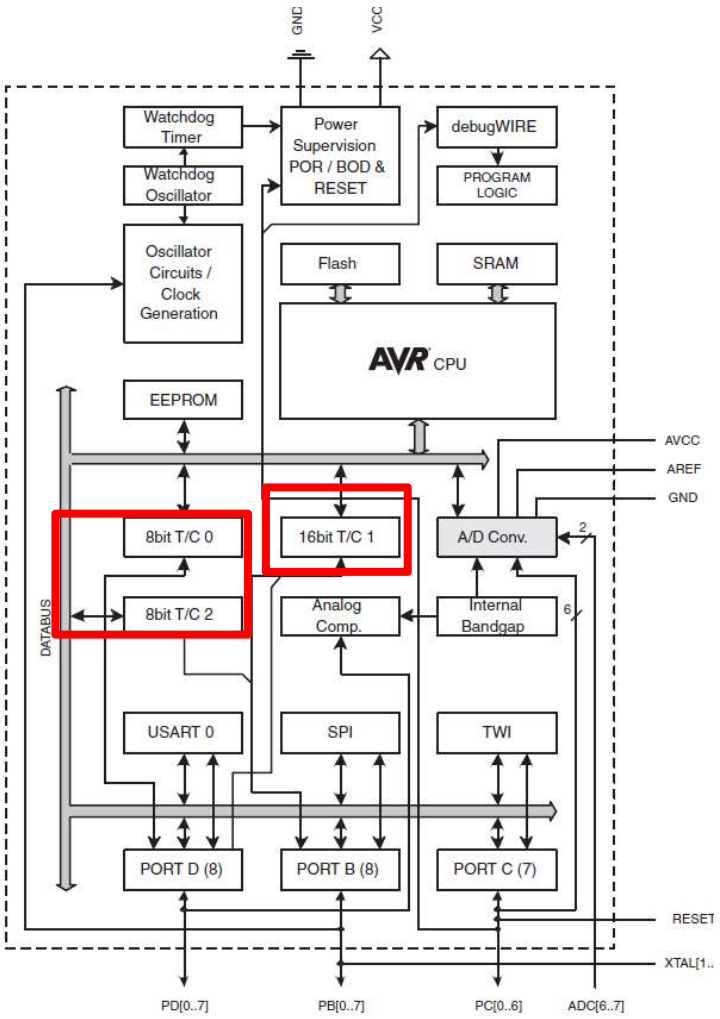
Timers

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DE SISTEMAS Y AUTOMÁTICA

Contadores

- ATmega328p dispone de dos contadores (timer/counter) de 8 bits y uno de 16.



Contadores

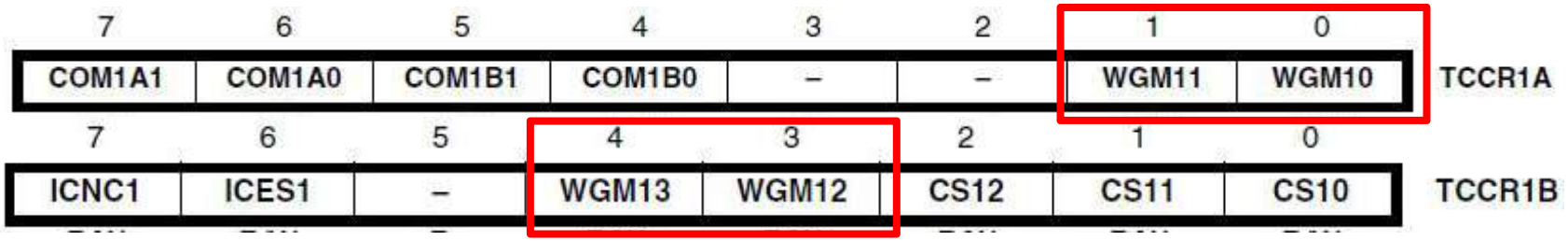
- ATmega328p contempla 16 modos de operación.

Mode	WGM13	WGM12 (CTC1)	WGM11 (PWM11)	WGM10 (PWM10)	Timer/Counter Mode of Operation	TOP	Update of OCR1x at	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	BOTTOM	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	BOTTOM	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	BOTTOM	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	(Reserved)	–	–	–
14	1	1	1	0	Fast PWM	ICR1	BOTTOM	TOP
15	1	1	1	1	Fast PWM	OCR1A	BOTTOM	TOP

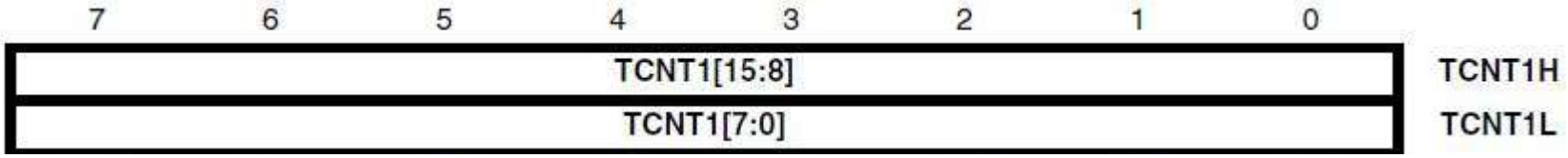
Contadores

Configuración del Timer/Counter 1 en modo normal

- Dos registros de configuración **T**imer/**C**ounter **C**ontrol **R**egisters **A**/**B** (TCCR1A y TCCR1B).



- La cuenta se realiza en los registros (TCNT1H y TCNT1L)

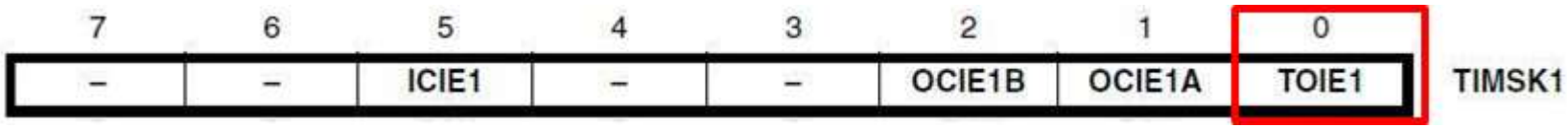


- TOV1 del registro TIFR1 (Interrupt flag register) se activa cuando se produce un overflow en la cuenta, comenzando una nueva cuenta desde 0x0000.



Contadores

- Los temporizadores generarán interrupciones si habilitamos sus máscaras individuales. Por ejemplo para activar la interrupción por overflow del Timer1:



y la ISR correspondiente será:

```
ISR(TIMER1_OVF_vect)
{
        //Do something
}
```

Contadores

Configuración del Timer/Counter 1 en modo normal

- La entrada del reloj (CLK) puede estar pre-escalada (dividida) por 1,8,64,256 y 1024.

CS12	CS11	CS10	Description
0	0	0	No clock source (Timer/Counter stopped).
0	0	1	$clk_{I/O}$/1 (No prescaling)
0	1	0	$clk_{I/O}$/8 (From prescaler)
0	1	1	$clk_{I/O}$/64 (From prescaler)
1	0	0	$clk_{I/O}$/256 (From prescaler)
1	0	1	$clk_{I/O}$/1024 (From prescaler)
1	1	0	External clock source on T1 pin. Clock on falling edge.
1	1	1	External clock source on T1 pin. Clock on rising edge.

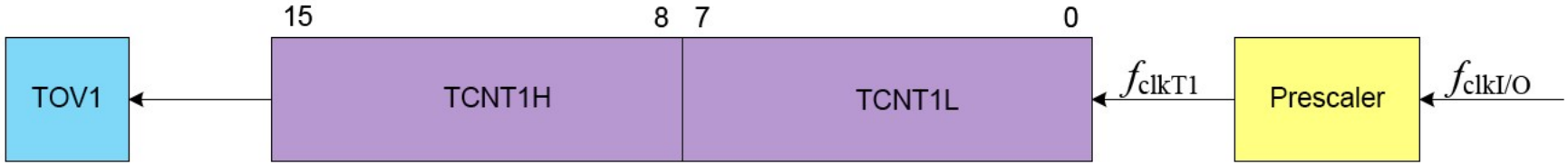
7	6	5	4	3	2	1	0	
ICNC1	ICES1	–	WGM13	WGM12	CS12	CS11	CS10	TCCR1B

Contadores

Configuración del Timer/Counter 1 en modo normal

RESUMEN

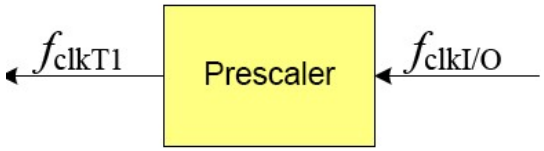
	7	6	5	4	3	2	1	0	
TCCR1A	COM1A1	COM1A0	COM1B1	COM1B0	–	–	WGM11	WGM10	
	R/W	R/W	R/W	R/W	R	R	R/W	R/W	
	0	0	0	0	0	0	0	0	
TCCR1B	ICNC1	ICES1	–	WGM13	WGM12	CS12	CS11	CS10	
	R/W	R/W	R	R/W	R/W	R/W	R/W	R/W	
	0	0	0	0	0	0	0	0	
TIFR1	–	–	ICF1	–	–	OCF1B	OCF1A	TOV1	
	R	R	R/W	R	R	R/W	R/W	R/W	
	0	0	0	0	0	0	0	0	



Contadores

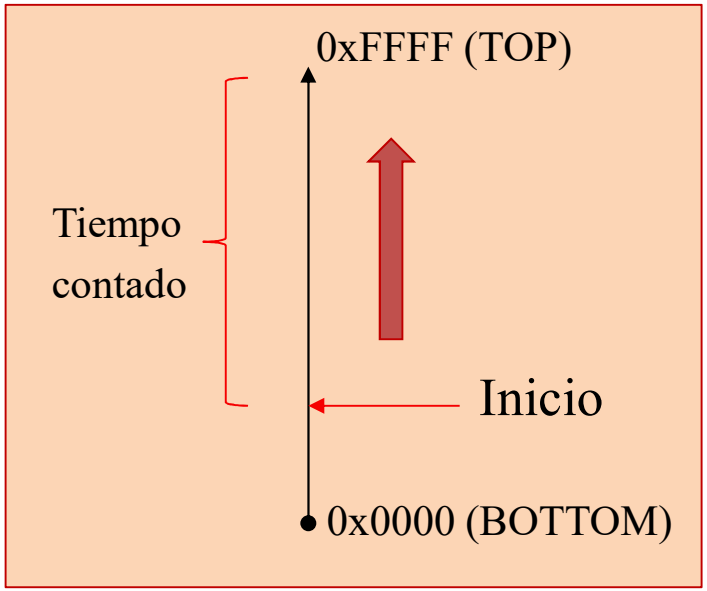
Ejemplo

- Se desea realizar una espera de 250 ms. considerando un reloj de 16 MHz y un escalado de 64 utilizando el contador de 16 bits.



$$f_{clkT1=}\frac{16 * 10^6 Hz}{64} = 250KHz$$

$$C=\frac{250 * 10^{-3}}{\frac{1}{250 * 10^3}} = 250 * 250 = 62500 < 2^{16}$$



- Palabra a cargar en TCNT1: $2^{16} - 62500 = 65536 - 62500 = 3036$ (Inicio de cuenta; hay que establecerlo cada vez que se quiera contar).
- 3036 =0x0BDC → TCNT1H=0x0B y TCNT1L=0xDC

Contadores

Configuración del Timer/Counter 0 en modo CTC

Mode	WGM02	WGM01	WGM00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on[(1)(2)]
0	0	0	0	Normal	0xFF	Immediate	MAX
1	0	0	1	PWM, Phase Correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	Immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	–	–	–
5	1	0	1	PWM, Phase Correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	–	–	–
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

- Dos registros de configuración **T**imer/**C**ounter **C**ontrol **R**egisters **A**/**B** (TCCR0A y TCCR0B).

7	6	5	4	3	2	1	0	
COM0A1	COM0A0	COM0B1	COM0B0	–	–	WGM01	WGM00	TCCR0A

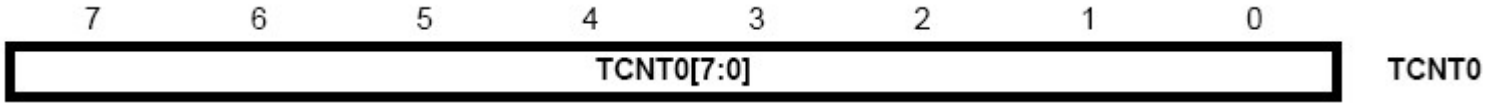
7	6	5	4	3	2	1	0	
FOC0A	FOC0B	–	–	WGM02	CS02	CS01	CS00	TCCR0B

Contadores

Configuración del Timer/Counter 0 en modo CTC

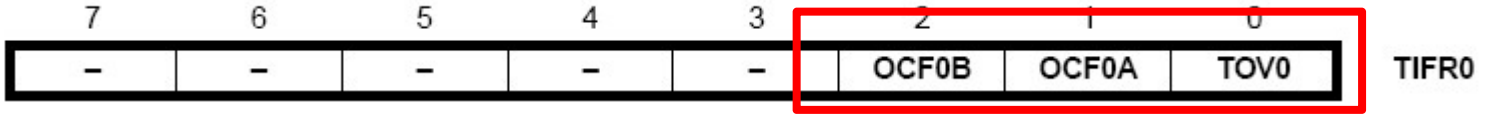
-

La cuenta se realiza en el registro (TCNT0)

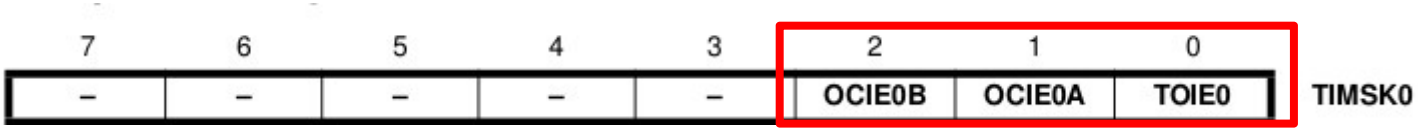


- Comparación de cuenta con los registros OCR0A[7:0] y/o OCR0B[7:0]

- El bit TOV0 del registro TIFR0 (Interrupt flag register) se activa cuando se produce un overflow en la cuenta; OCF0A cuando la cuenta alcance el valor de OCR0A y OCF0B cuando se alcance el valor de OCR0B

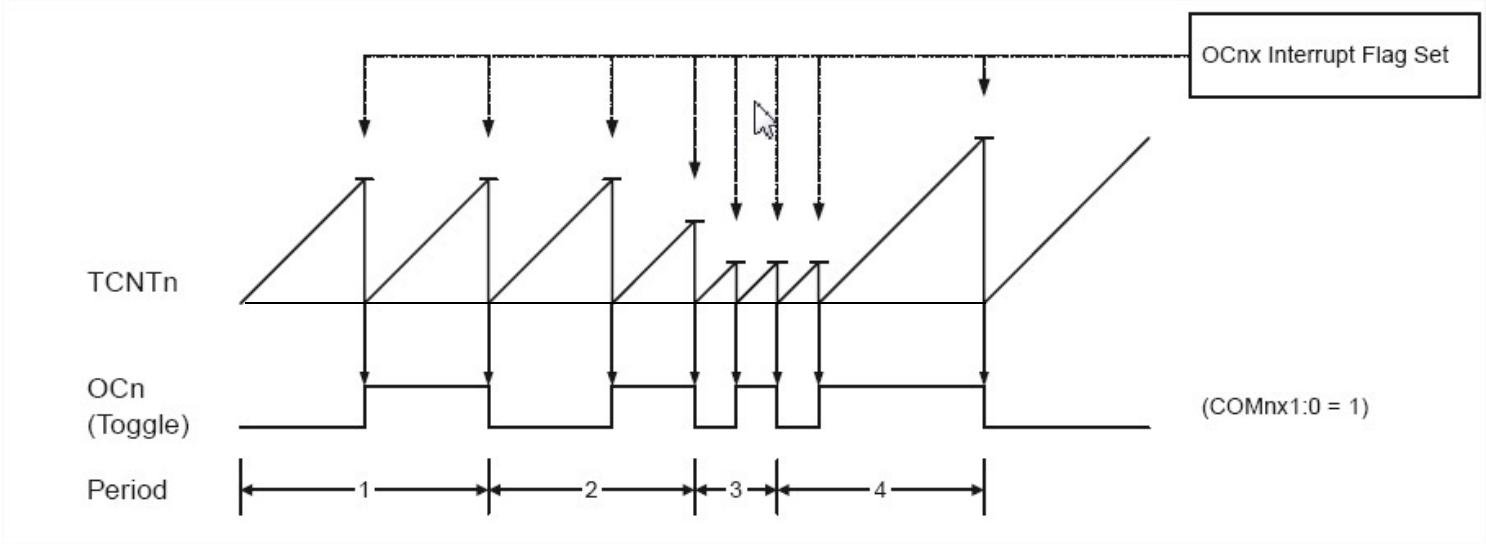


- Y se generarán interrupciones si se habilitan en la mascara correspondiente



Contadores

Configuración del Timer/Counter 0 en modo CTC



7	6	5	4	3	2	1	0	
COM0A1	COM0A0	COM0B1	COM0B0	–	–	WGM01	WGM00	TCCR0A

COM0A1	COM0A0	Description
0	0	Normal port operation, OC0A disconnected.
0	1	Toggle OC0A on Compare Match
1	0	Clear OC0A on Compare Match
1	1	Set OC0A on Compare Match

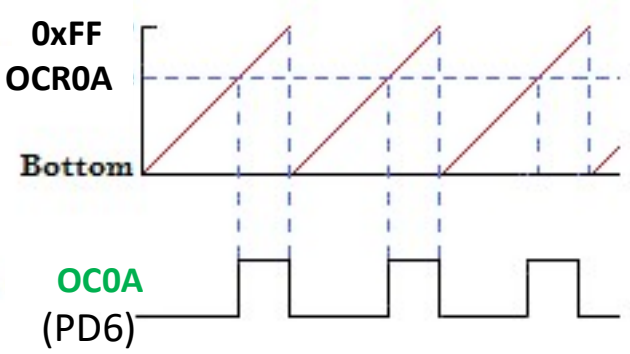
OC0A→PD6

PWM

Configuración del Timer/Counter 0 en modo FastPWM

Mode	WGM02	WGM01	WGM00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on[(1)(2)]
0	0	0	0	Normal	0xFF	Immediate	MAX
1	0	0	1	PWM, Phase Correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	Immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	–	–	–
5	1	0	1	PWM, Phase Correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	–	–	–
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

Cuenta hasta **0xFF (TOP)** y utiliza **OCR0A (Compare)** para fijar el duty cycle



Modo Inverted:

OC0A=1 cuando el timer alcanza el valor de OCR0A cuando cuenta hacia arriba

OC0A=0 cuando se comienza en Bottom.

PWM

Configuración del Timer/Counter 0 en modo FastPWM

COM0A1	COM0A0	Description
0	0	Normal port operation, OC0A disconnected.
0	1	WGM02 = 0: Normal Port Operation, OC0A Disconnected. WGM02 = 1: Toggle OC0A on Compare Match.
1	0	Clear OC0A on Compare Match, set OC0A at BOTTOM, (non-inverting mode).
1	1	Set OC0A on Compare Match, clear OC0A at BOTTOM, (inverting mode).

7	6	5	4	3	2	1	0	
COM0A1	COM0A0	COM0B1	COM0B0	–	–	WGM01	WGM00	TCCR0A

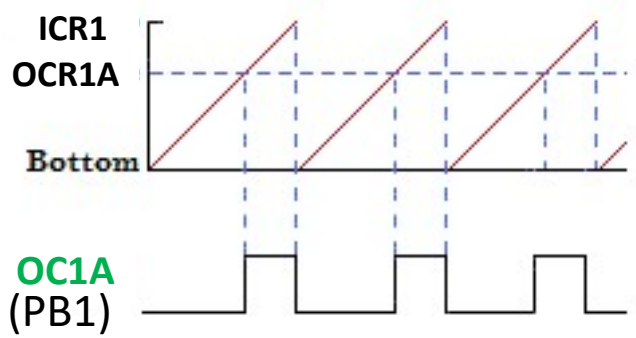
¿Qué obtendríamos si configuramos el **modo non-inverting** (COM0A1=1 & COM0A0=0)?

PWM

Configuración del Timer/Counter 1 en modo FAST PWM

Mode	WGM13	WGM12 (CTC1)	WGM11 (PWM11)	WGM10 (PWM10)	Timer/Counter Mode of Operation	TOP	Update of OCR1x at	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	BOTTOM	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	BOTTOM	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	BOTTOM	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	(Reserved)	–	–	–
14	1	1	1	0	Fast PWM	ICR1	BOTTOM	TOP
15	1	1	1	1	Fast PWM	OCR1A	BOTTOM	TOP

Cuenta hasta **ICR1 (TOP)** y utiliza **OCR1A (Compare)** para fijar el duty cycle



PWM

Configuración del Timer/Counter1 en modo FastPWM

COM1A1/COM1B1	COM1A0/COM1B0	Description
0	0	Normal port operation, OC1A/OC1B disconnected.
0	1	WGM13:0 = 14 or 15: Toggle OC1A on Compare Match, OC1B disconnected (normal port operation). For all other WGM1 settings, normal port operation, OC1A/OC1B disconnected.
1	0	Clear OC1A/OC1B on Compare Match, set OC1A/OC1B at BOTTOM (non-inverting mode)
1	1	Set OC1A/OC1B on Compare Match, clear OC1A/OC1B at BOTTOM (inverting mode)

COM1A1	COM1A0	COM1B1	COM1B0	–	–	WGM11	WGM10	TCCR1A